Thadomal Shahani Engineering College, Bandra
Department of Artificial Intelligence & Data Science

Name: Chaitanya Vijay Jhade				**Subject: ML Laboratory**
TE- 1	**Batch: T13**	**Sem: 6**	**Roll No: 2301048**	**Date: 19-01-2026**

Experiment No: 02

Aim: To understand Version Control System / Source Code Management, install git and create a GitHub account

Theory:

- Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

- Some of the basic operations in Git are:

    1. Initialize
    2. Add
    3. Commit
    4. Pull
    5. Push

- Some advanced Git operations are:
    1. Branching
    2. Merging
    3. Rebasing

The following diagram depict the all supported operations in GIT

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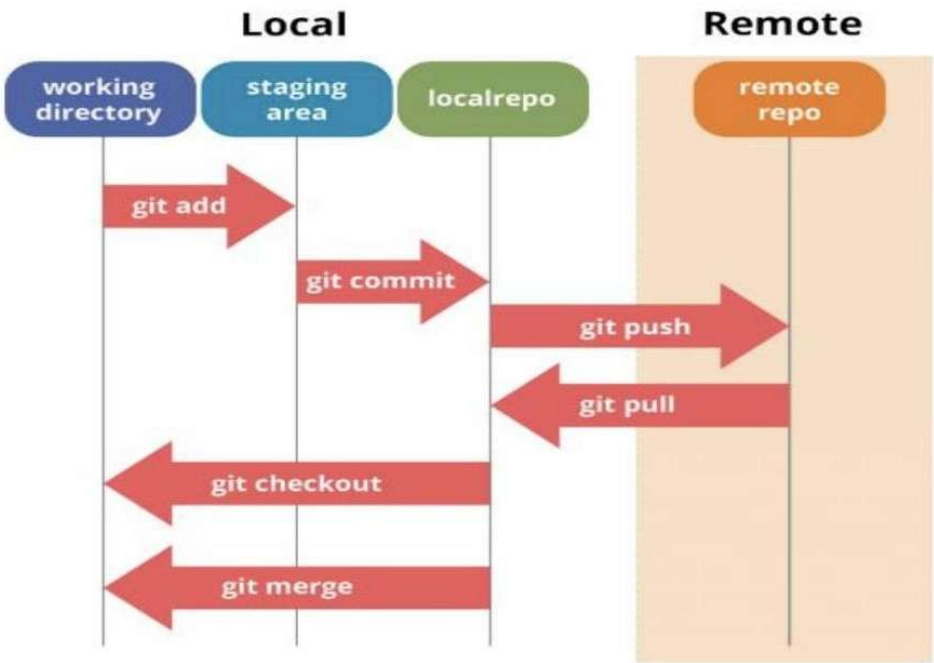

- **Installation of GIT**

1) In windows, download GIT from https://git-scm.com/ and perform the straightforward installation.

2) In Ubuntu, install GIT using $sudo apt install git,

Confirm the version after installation using command $git –version

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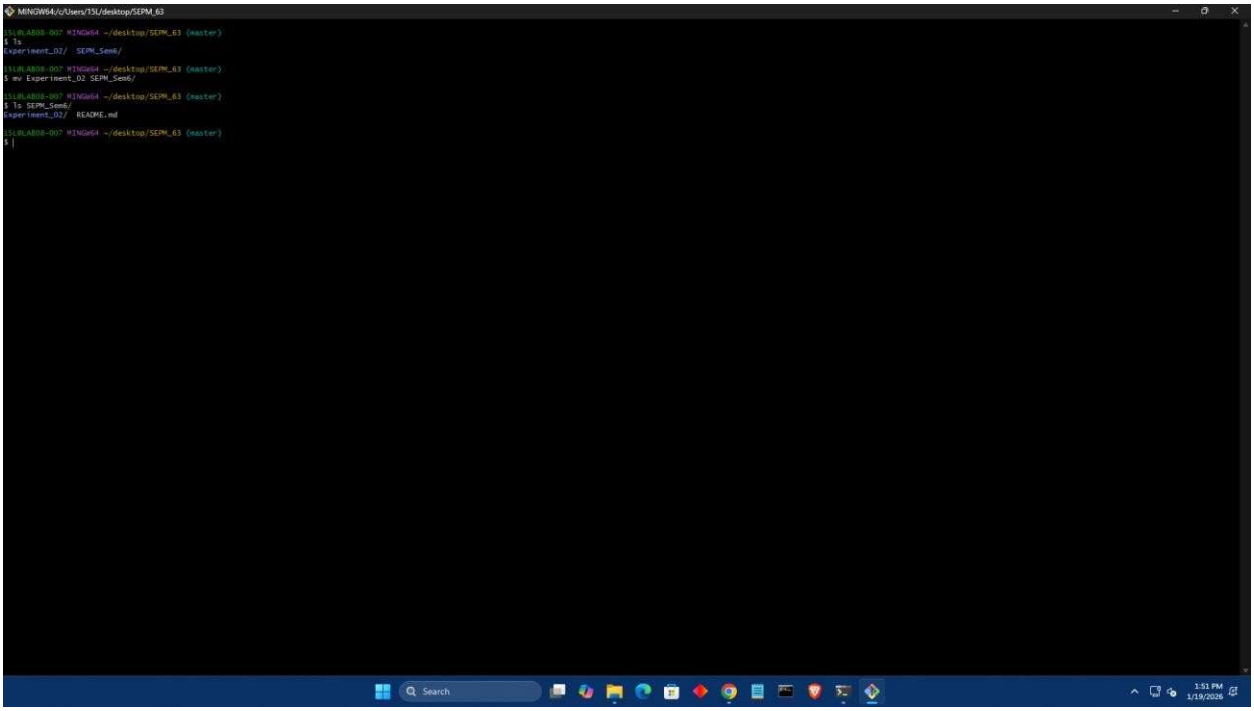
Output:



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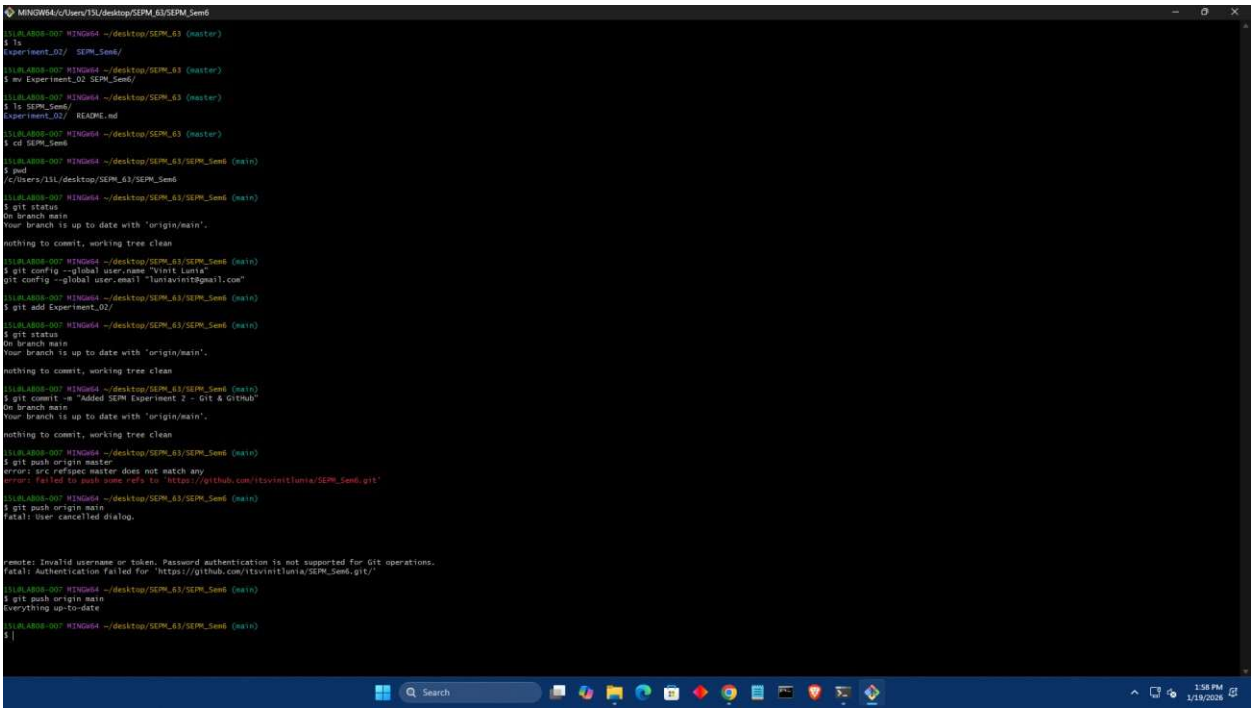

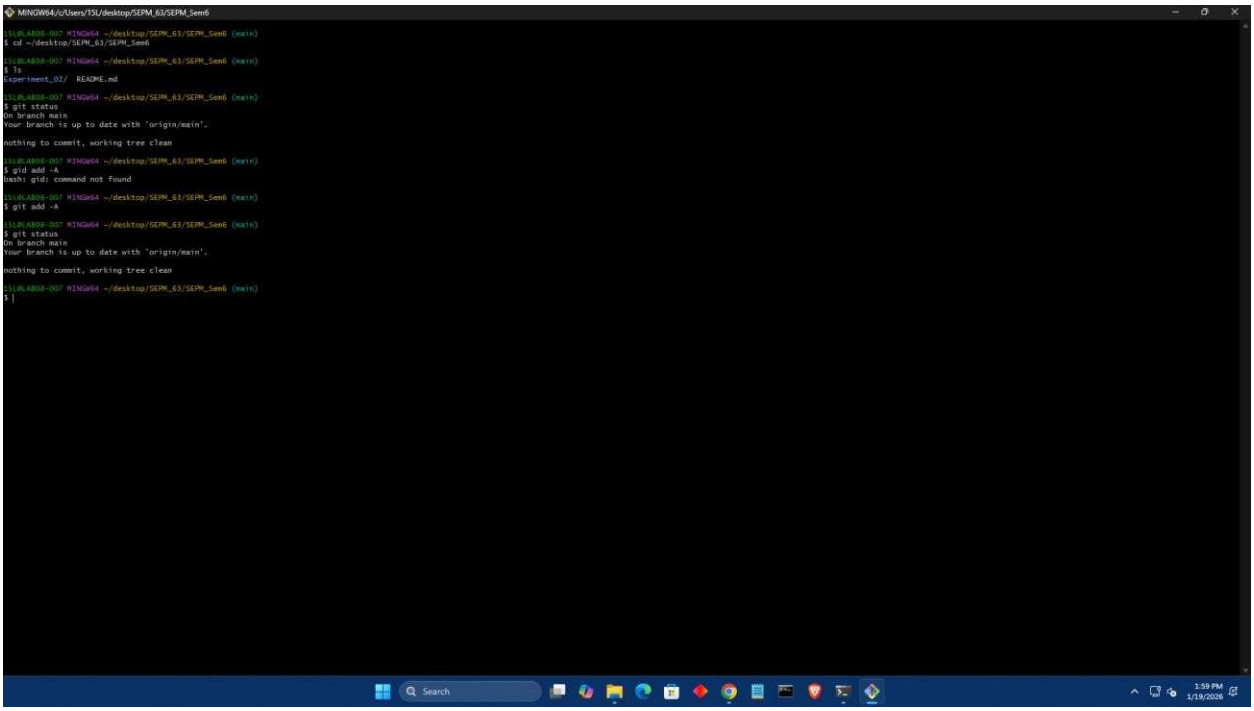

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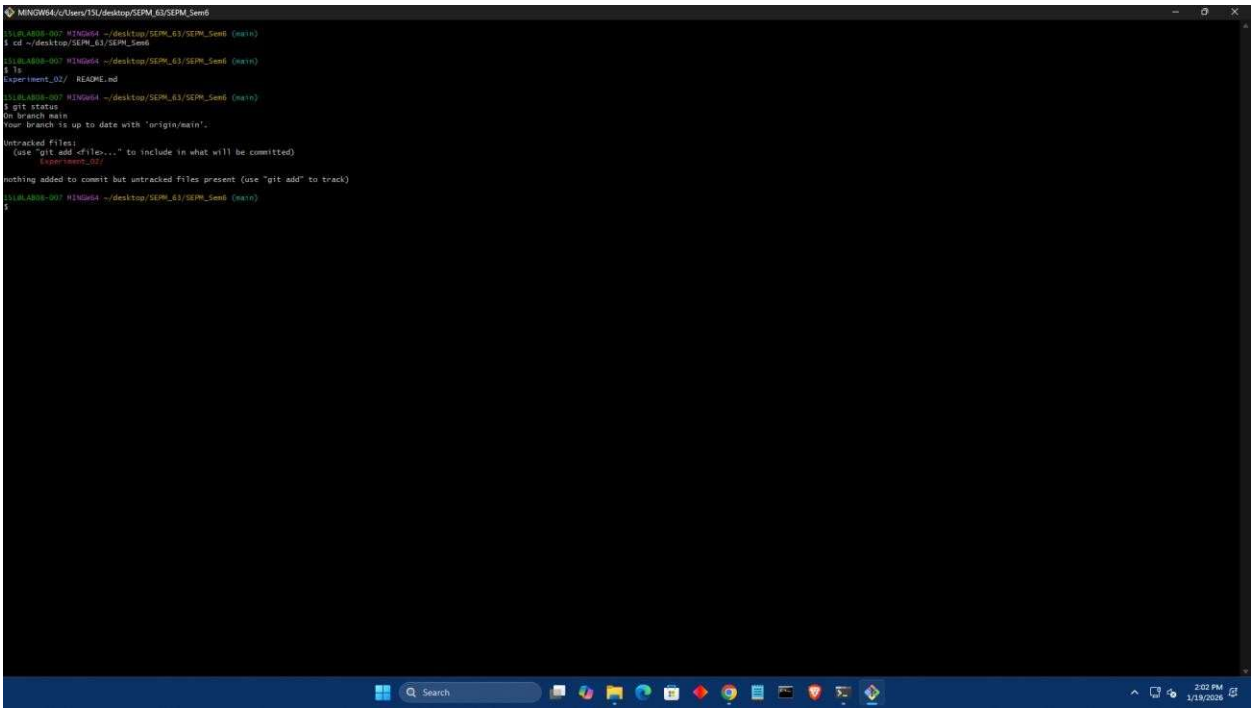

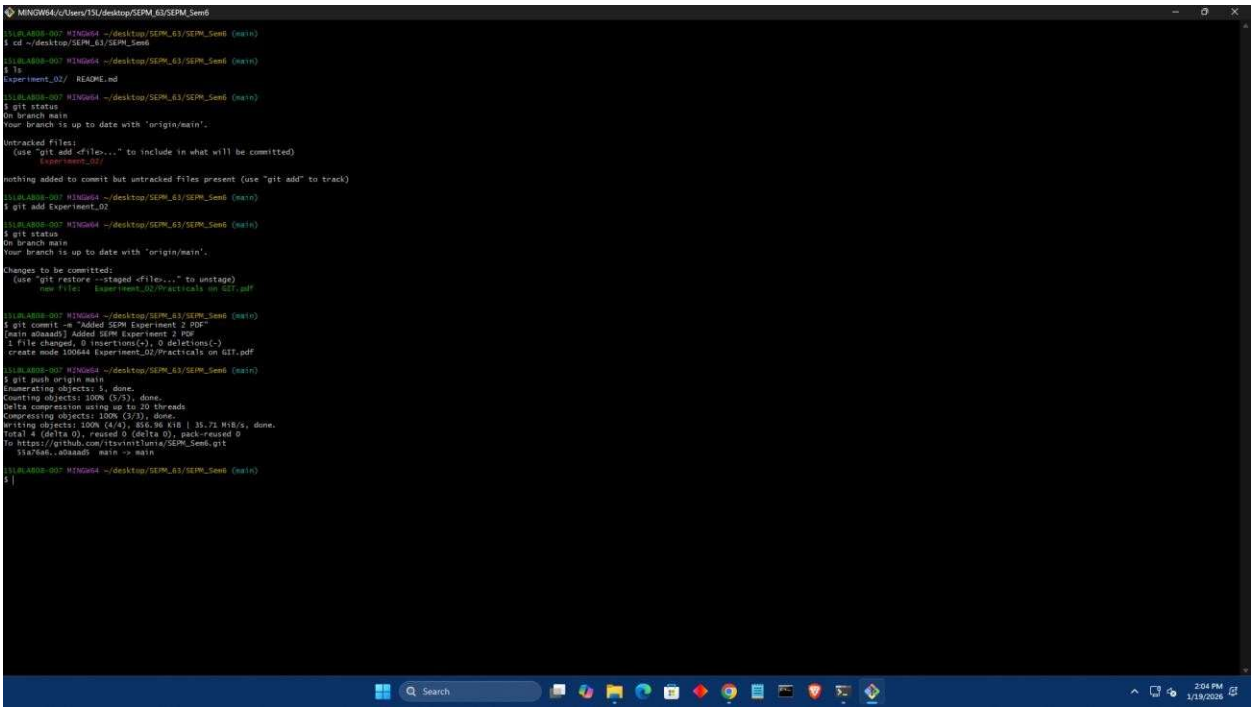

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Conclusion: Successfully performed Version Control System / Source Code Management, install git and create a GitHub account